

Hybrid Classical-Quantum Models for Real World Applications

This project develops a foundational framework for integrating quantum machine learning (QML) with classical ML/DL pipelines for applications. Using publicly available datasets as the initial case study, you will build hybrid models that combine parameterized quantum circuits (PQCs) with established deep learning architectures. The goal is to evaluate where and how quantum components can enhance classical baselines in terms of predictive performance, feature representation, and model interpretability. This plan progresses through three phases: foundational learning and literature review, algorithm development and implementation, and validation with documentation. Upon completion, you will aim to have working hybrid QML prototypes with benchmark comparisons against classical methods.

Quantum machine learning offers a new paradigm for data analytics by encoding classical data into quantum states, potentially revealing latent relationships inaccessible to classical models. Hybrid quantum-classical workflows are becoming increasingly practical as quantum hardware evolves, with integration of quantum circuits into classical computing and HPC pipelines viewed as the most viable pathway for near-term applications.

1. Quantum feature encodings: Using quantum kernels and feature maps to provide richer class separability, with metrics like normalized global effective dimension (NGED) to assess trainability and capacity.

Project Timeline

[illegible]

Phase 2: Algorithm Design and Development

Building on Phase 1 foundations, this phase focuses on implementing hybrid quantum-classical models and establishing classical baselines for comparison.

Week 3-5: Classical Baseline Implementation

Objectives

- Implement strong classical baselines for cardiovascular disease prediction
- Prepare benchmark datasets (structured clinical data, imaging)
- Establish evaluation metrics and validation protocols

Key Activities

Data Preparation: Process publicly available datasets. Implement preprocessing pipelines for structured data normalization and image augmentation.

Classical Models: Implement baseline classifiers including Random Forest and XGBoost for structured data, DenseNet-121 or ResNet for chest X-ray classification, and Logistic regression and SVM baselines.

Evaluation Framework: Set up evaluation pipeline with ROC AUC, accuracy, precision, recall, F1, and calibration metrics (ECE).

Deliverables

1. Preprocessed benchmark datasets ready for experimentation
2. Classical baseline models with documented performance metrics

Week 5-9: Hybrid QML Implementation

Objectives

- Implement parameterized quantum circuits for classification tasks
- Develop hybrid NN+PQC architecture
- Implement quantum kernel methods for structured data
- Compare different quantum feature maps and ansätze

Key Activities

Quantum Circuit Design: Build parameterized quantum circuits using Qiskit: implement angle and amplitude encoding schemes, design variational ansätze (hardware-efficient, problem-inspired), create quantum kernel computation pipelines.

Hybrid Model: Develop hybrid architecture combining NN, dimensionality reduction layer, PQC-based classifier head. Train on subset.

Quantum Kernel SVM: Implement quantum kernel methods for structured clinical data: ZZFeatureMap and other quantum feature maps, fidelity-based kernel computation, integration with classical SVM.

Interpretability Integration: Implement Grad-CAM++ for hybrid imaging models to visualize which image regions drive predictions.

Deliverables

1. Working hybrid QML implementations (NN+PQC and QSVM)
2. Comparison of quantum encoding strategies with performance metrics

Phase 3: Testing, Optimization, and Documentation

The final phase focuses on rigorous validation, benchmarking against classical baselines, and comprehensive documentation.

Week 9-11: Validation and Benchmarking

Objectives

- Systematically compare hybrid QML models against classical baselines
- Evaluate trainability using normalized global effective dimension (NGED)
- Assess model calibration and uncertainty quantification
- Conduct ablation studies to isolate quantum component contributions

Key Activities

Performance Comparison: Compare hybrid models against classical baselines on classification metrics (ROC AUC, F1, accuracy), calibration (expected calibration error), and training stability and convergence.

Quantum Component Analysis: Compute NGED to assess quantum circuit expressibility, conduct ablation studies removing/replacing quantum layers, analyze noise impact using simulated device noise.

Interpretability Evaluation: Compare saliency map quality between hybrid and classical models for imaging tasks.

Deliverables

1. Comprehensive benchmark report with statistical comparisons
2. Ablation study results quantifying quantum component contribution

Week 10-12: Documentation and Final Report

Objectives

- Prepare comprehensive documentation including tutorials
- Draft technical report summarizing findings
- Package code for reproducibility

Key Activities

Code Documentation: Document all modules with docstrings and usage examples, create Jupyter notebook tutorials for key workflows, set up reproducible environment (requirements.txt, Docker).

Technical Report: Draft report covering background and motivation, methods and implementation details, results and benchmark comparisons, discussion of when QML adds value, and future directions.

Repository Preparation: Organize GitHub repository with clean, modular code structure, README with installation and usage instructions, example notebooks, benchmark results and figures.

Deliverables

1. Complete GitHub repository with documented code
2. Draft technical report ready for review
3. Tutorial notebooks demonstrating hybrid QML workflows

Suggested Reads:

1. Abbas, A., et al. (2021). The power of quantum neural networks. *Nature Computational Science*, 1(6), 403-409.
2. Schuld, M., & Petruccione, F. (2021). *Machine Learning with Quantum Computers*. Springer.
3. Huang, H.Y., et al. (2021). Power of data in quantum machine learning. *Nature Communications*, 12, 2631.